

Guardians of The Galaxy Vol 3: The Creation and Destruction of The Arête

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ABSTRACT

Guardians of the Galaxy Vol. 3 is Director James Gunn's epic final instalment in Marvel Studios' Guardians series, where we're introduced to old enemies, bigger spaceships and explosive battles. Wētā FX Sequence VFX Supervisor Jason Galeon will share how the team constructed the most complex spaceship model they've ever created with detailed sub environments using new workflows.

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1 INTRODUCTION

The ruby red Arête Spacecraft is featured at the centre of much of the film's action. Measuring 4 kilometer wide, it provided a range of sub-environments to stage many of the story beats. Character moments and dynamic fight sequences take place all over and inside the Arête, requiring it to be constructed in a highly modular and art-directable manner. This talk will cover how a singular modelling approach was used to design the spacecraft for use in wide master shots in outer space, detailed close-up shots on counter-earth, as well as a glorious 150-second-long fight sequence inside the ship's spaceport.

2 PRODUCTION

This design process began with breaking down the concept art. The colossal ship would be seen from afar in space, as well as in extreme close-up, with the camera falling mere feet away from the surface. The Arête had intricate panelling and mechanics visible both on its surface and deep behind a ruby glass façade and set in a harbour surrounded by waterfalls.

To construct a structure of this scope the team approached it less like a building, and more like a city, creating a set of clearly defined rules and procedural logic to generate the layout before developing in more bespoke detail. To achieve this, a new workflow was developed to increase the creative iterations and to add to the build as it was coming together. The workflow involved modelling in Maya, a Citybuilder approach to layout in Houdini, look development in Wētā FX's proprietary shading system SQUID, scene management

using USD, and rendering using Katana and Manuka. And, like most big model builds in VFX, the Arête was designed for destruction – with each level of the build modelled volumetrically and rigged to be blown up.

The exterior of the Arête was constructed using a combination of procedurally laid out geometry and individually modelled details, to ensure that any given region of the model maintained its own unique character and did not appear randomly generated. The Central Core was modelled piece by piece using custom sets of 'greeblies' (a term first used by ILM to describe the many small details on the death star model) to distinguish it from the glass sections. Various structural components like antennas, buildings, platforms, and vents sourced from a library of about 30 assets were built to create a coherent look to the core. A full-scale Spaceport was built into the Arête to provide a seamless set where the Guardians crash land to carry on the third act inside the spacecraft itself.

A large set of greeblies were created at an array of different scales. Some of these were to serve as structural elements, while others gave more sense of function to the workings of the Arête. Our Layout department created a new workflow to distribute these greeblies along the structure of the Arête based on a set of rules, like a 3-dimensional jigsaw puzzle. For example, in order to avoid intersections, the greeblies were distributed using the UV space of each panel. A silhouette of each asset was treated as a UV face and laid out using UV tiling. The assets were then swapped back onto each silhouette.

Different sections of the Arête required different levels of specific depth and structure. For these, the greeblies were laid out in grids or lines to form a cube of assets overloaded with other assets. This cube could then be tiled to fit the bespoke geometry of the Arête. To avoid repetition and using more assets than necessary, the greeblies could be culled from the tiled strip randomly to create unique more organic placement. These could be art directed both in how the greeblies were laid out in the original tile and how they were culled from the strip, both in distribution and volume, by adjusting a set of rules and logic.

We approached the design organically and procedurally because it was impractical to model every square inch of kilometres of surface area. For the ruby red glass surfaces cladding much of Arête, we created a Houdini tool that analyzes each surface and divides it into a grid in a preferred orientation. Then, generating curves from this grid to create a falloff mask, these panels would be subdivided vertically or horizontally based on random switches, with further octaves created on the fly based on the falloff from the mask. We could then extrude these faces into several layers based on size, iteration and noise.

To create the red glass shaders, we clad the Arête in a layer of metallic coating, adjusted the glass properties like index of refraction closer to ruby, and applied weathering to the surface. The Arête

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is composed of hundreds of unique sub components. To manage these myriad assets, we used nested USD layering for the first time in production at Wētā FX.

We also used our proprietary shader system, *Squid*, for the first time in production at Wētā FX. Squid constrains a set of tools that allows users to define a common shading graph, and generate as many resulting monolithic shaders out of it, one for each supported shading language.

The primary challenge in animating the Arête was finding a way to transfer the huge amount of geometry between departments. Our standard workflows quickly become unable to process an asset of this size and scale purely based on the RAM required to process the geometry. We leveraged our in-house constraint system for driving the components that combine to construct Arête. We constructed a puppet which allowed motion to be applied to all the various quadrants and bridging structures that made up Arête, with visualization of simplified low poly geometry making it performant for animation. Layout also leveraged this for visual reference to guide their construction process. Driving the hero geometry directly in the layout itself meant Arête could enter the lighting scenes as a large-scale layout that still received animated input. This saved a lot of costly geometry processing and enabled efficient turnaround times. An additional benefit to this was that hero details could continue to be added and adjusted without having to continually update the rigs - allowing work to happen simultaneously across departments in a non-destructive way.

As we were designing the Arête we were previsualizing this final battle in outer space, so the team had to prepare for any section of it to be seen from every possible angle. Carrying this amount

of data would normally break any pipeline, so we took a modular approach to the hierarchy of the model to be able to isolate only the parts we needed, which could range from the width of a house to the size of Seattle.

The final act of the film features the slow destruction of Arête from a chain reaction of explosions - as the giant skull of Knowhere links up to the deteriorating ship, allowing the heroes to escape.

This required the entire Arête model to be destruction-ready, with full internal structures designed and modelled to reveal the insides as it fractures apart.

FX Simulations were run on all parts of the Arête to make it explode from the inside and fracture apart into a myriad of pieces large and small.

Establishing scale and ensuring the action was legible presented a challenge in this destruction sequence, with pieces ranging from small shards of glass to fractured pieces of debris the size of football stadiums.

The talk will conclude by walking through how they worked backward from the bespoke destruction to ensure the build maintained design integrity and realism throughout.

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